ECON2103 Microeconomics

Chapter 8 Profit Maximization and Competitive Supply

Dr. Sherry Zhou



BEIJING NORMAL UNIVERSITY · HONG KONG BAPTIST UNIVERSITY
UNITED INTERNATIONAL COLLEGE

Outline

1. Perfectly Competitive Markets
2. Marginal Revenue, Marginal Cost, and Profit Maximization
3. Choosing Output in the Short Run
4. Competitive Firm's Short-Run Supply Curve
5. The Short-Run Market Supply Curve
6. Choosing Output in the Long Run
7. The Industry's Long-Run Supply Curve
8. Summary and Exercises

Three Basic Assumptions: Assumption 1

The model of perfect competitive rests on three basic assumptions:

1. **Price Taking**
   - Because each individual firm sells a sufficiently small proportion of total market output, its decisions have no impact on market price.
   - **price taker** - Firm that has no influence over market price and thus takes the price as given.

Three Basic Assumptions: Assumption 2

2. **Product Homogeneity**
   - When the products of all of the firms in a market are perfectly substitutable with one another—that is, when they are homogeneous—no firm can raise the price of its product above the price of other firms without losing most or all of its business.
   - In contrast, when products are heterogeneous, each firm has the opportunity to raise its price above that of its competitors without losing all of its sales.

Three Basic Assumptions: Assumption 3

3. **Free Entry and Exit**
   - **free entry (or exit)** - Condition under which there are no special costs that make it difficult for a firm to enter (or exit) an industry.
   - Buyers can easily switch from one supplier to another, and suppliers can easily enter or exit a market.

When Is a Market Highly Competitive?

- Many markets are *highly* competitive in the sense that firms face highly elastic demand curves and relatively easy entry and exit.
- Because firms can implicitly or explicitly collude in setting prices, the presence of many firms is not sufficient for an industry to approximate perfect competition.
- Conversely, the presence of only a few firms in a market does not rule out competitive behavior.

Profit Maximization

Do Firms Maximize Profit?

- The assumption of *profit maximization* predicts business behavior reasonably accurately and avoids unnecessary analytical complications.
- For smaller firms, profit is likely to dominate almost all decisions.
- In larger firms, however, managers may be more concerned with such goals as revenue maximization, revenue growth.
- Firms that do not come close to maximizing profit are not likely to survive. The firms that do survive make long-run profit maximization one of their highest priorities.

Marginal Revenue

- **profit** - Difference between total revenue and total cost.

$$\pi(q) = R(q) - C(q)$$

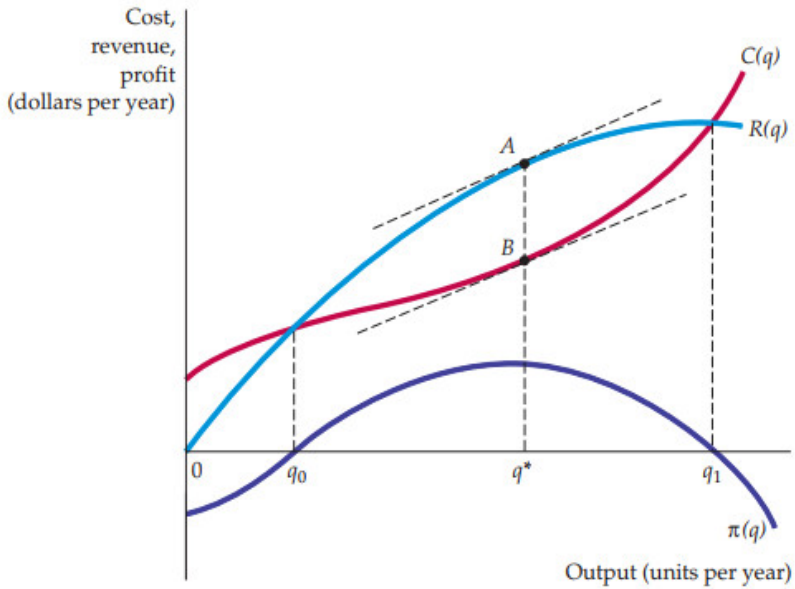
- **marginal revenue** - Change in revenue resulting from a one-unit increase in output.

Profit Maximization in the Short Run

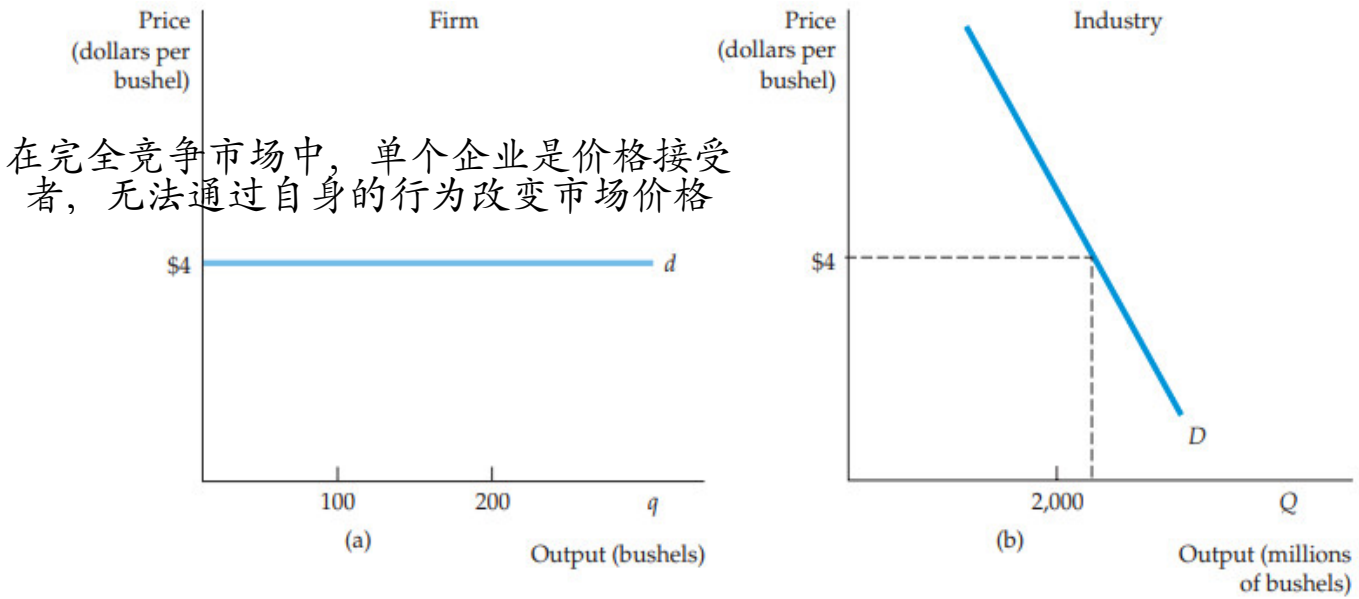
A firm chooses output $q*$ so that profit is maximized. At that point, marginal revenue is equal to marginal cost.

$$\Delta\pi/\Delta q = \Delta R/\Delta q - \Delta C/\Delta q = 0$$

$$MR(q) = MC(q)$$



Demand and Marginal Revenue for a Competitive Firm



- In (a), the demand curve facing the firm is perfectly elastic.
- In (b), the market demand curve is downward sloping.

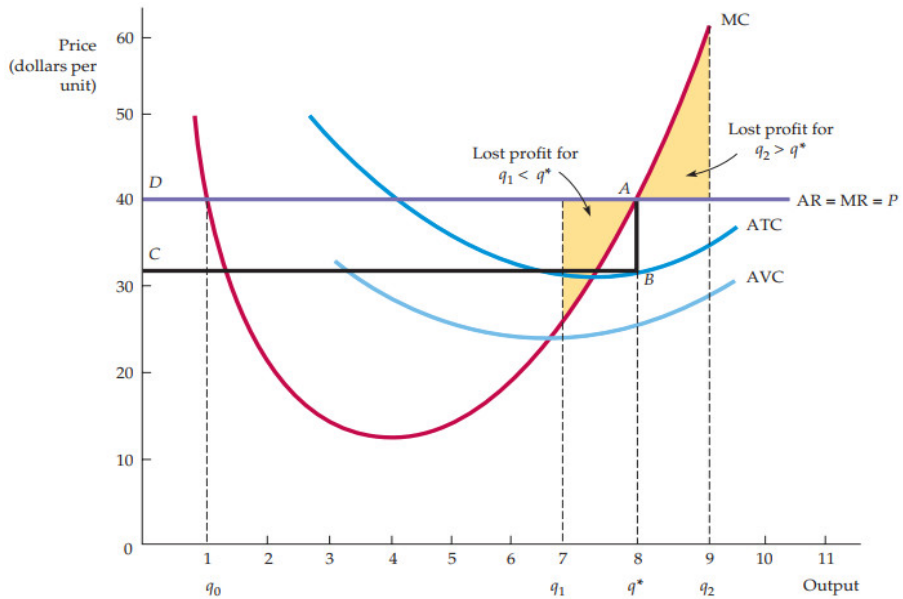
Short-Run Profit Maximization by a Competitive Firm

- The demand curve d facing an individual firm in a competitive market is both its average revenue curve and marginal revenue curve. Along this demand curve, marginal revenue, average revenue, and price are all equal.
- A perfectly competitive firm should choose its output so that marginal cost equals price:

$$MC(q) = MR = P$$

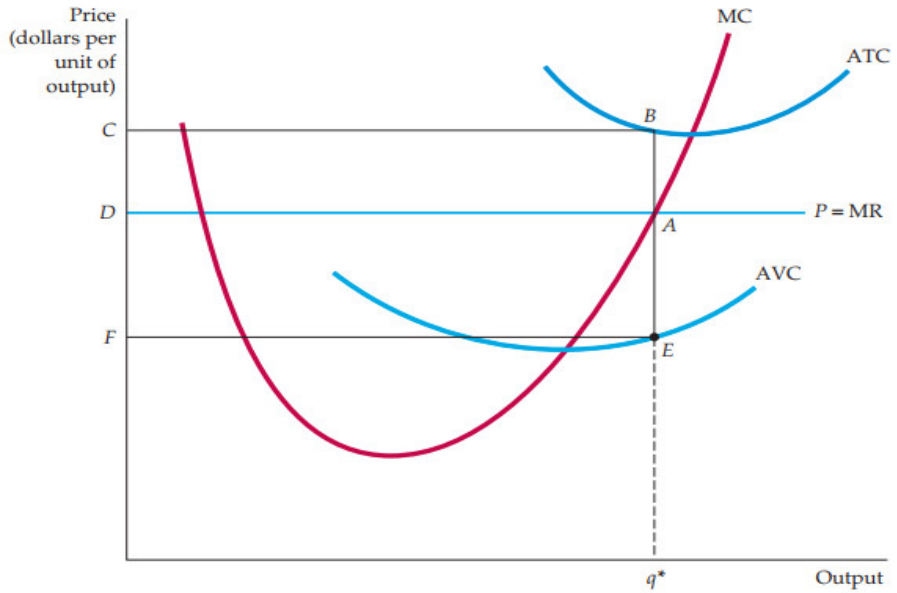
在完全竞争市场中：单个企业对市场价格没有任何影响。
每销售一单位产品的收益与价格一致（因为不能改变价格）。
所以，单个企业的需求曲线也是其平均收益和边际收益曲线，且这些值始终等于市场价格。

Short-Run Profit Maximization by a Competitive Firm



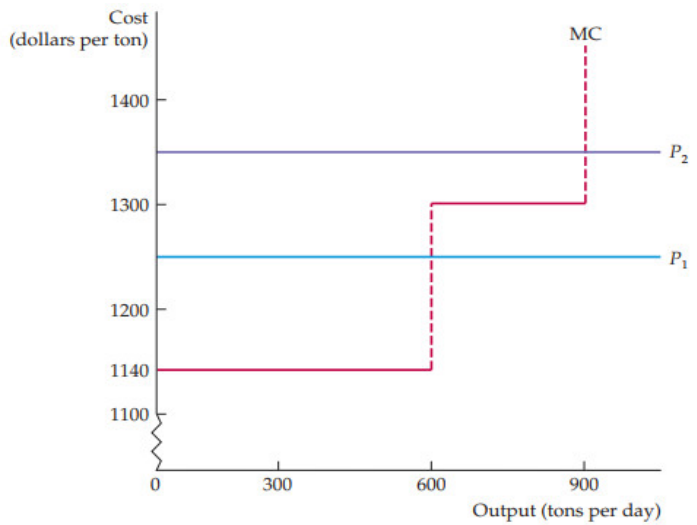
- **Output Rule**: If a firm is producing any output, it should produce at the level at which marginal revenue equals marginal cost.
- The profit of the firm is measured by the rectangle *ABCD*.

When Should the Firm Shut Down?



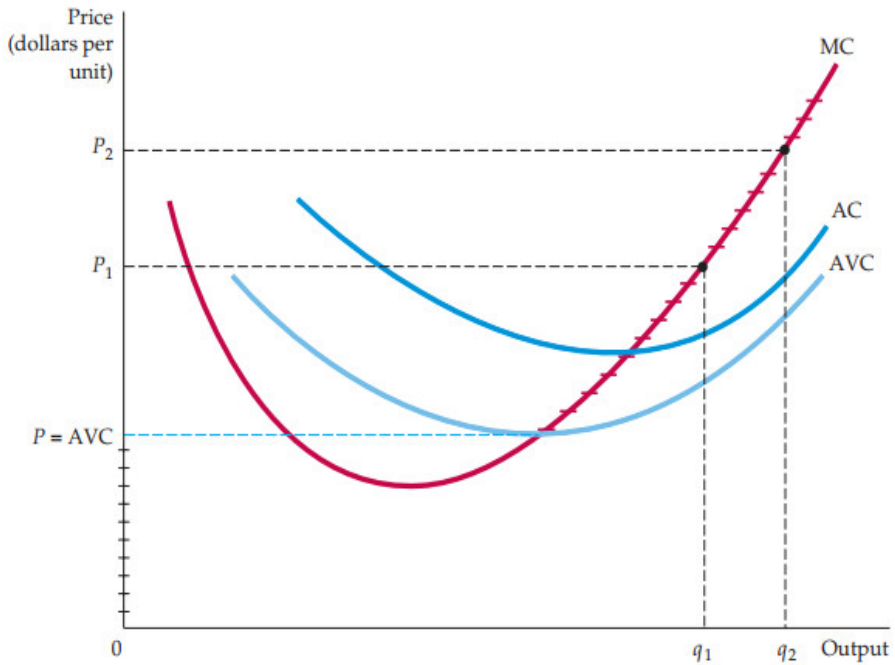
- A competitive firm should shut down if price is below AVC.
- The firm may produce in the short run if price is greater than AVC.

Example: The Short-Run Output Decision



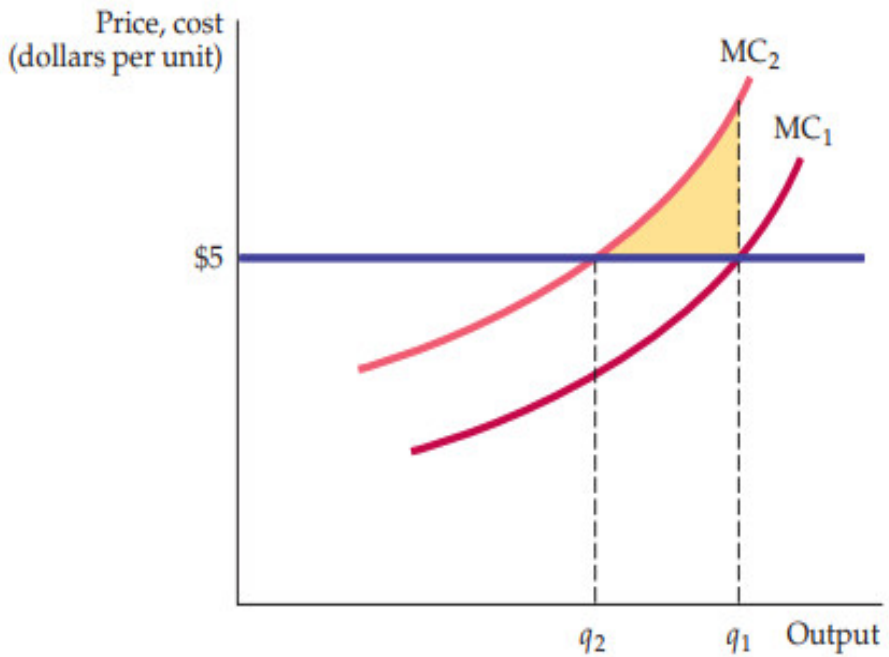
- In the short run, the plant should produce 600 tons per day if price is above \$1140 per ton but less than \$1300 per ton.
- If price is greater than \$1300 per ton, it should run an overtime shift and produce 900 tons per day.
- If price drops below \$1140 per ton, the firm should stop producing, but it should probably stay in business because the price may rise in the future.

The Competitive Firm's Short-Run Supply Curve



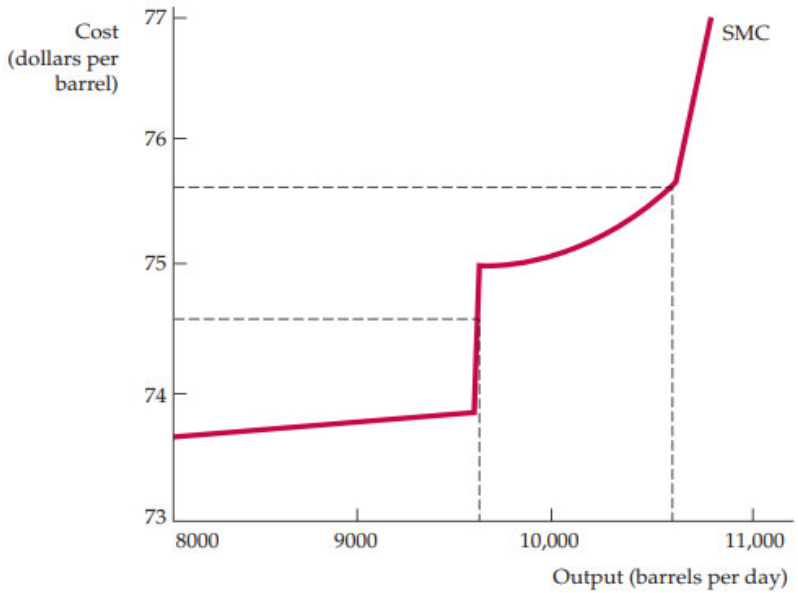
The firm's supply curve is the portion of the marginal cost curve for which marginal cost is greater than average variable cost.

The Firm's Response to an Input Price Change



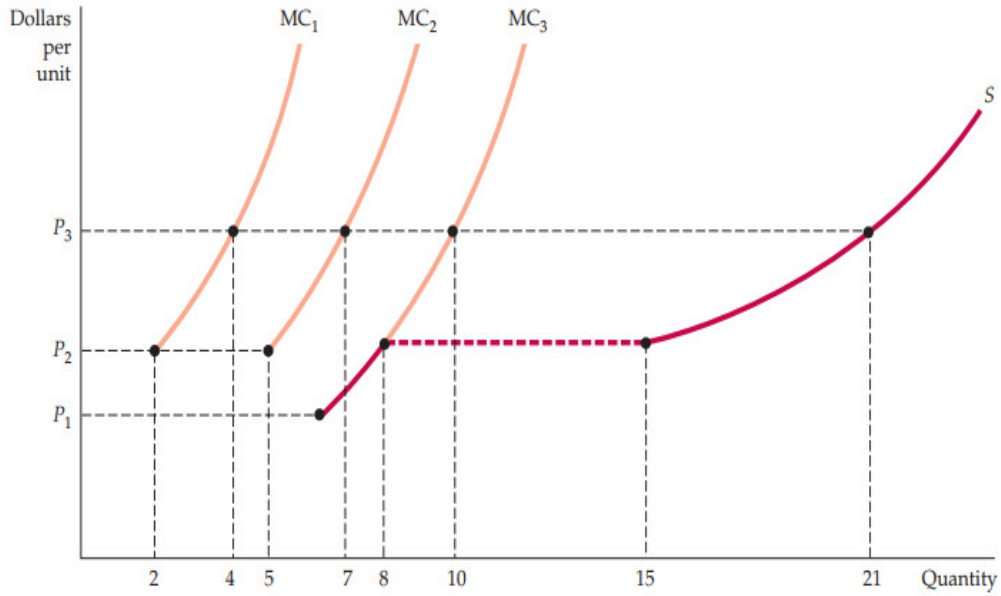
When the marginal cost of production for a firm increases (from MC_1 to MC_2), the level of output that maximizes profit falls (from q_1 to q_2).

Example: Short-Run Production of Petroleum Products



- As the refinery shifts from one processing unit to another, the marginal cost of producing petroleum products from crude oil increases sharply at several levels of output.
- As a result, the output level can be insensitive to some changes in price but very sensitive to others.

Industry Supply in the Short Run



The short-run industry supply curve is the summation of the supply curves of the individual firms.

Elasticity of Market Supply

$$E_S = (\Delta Q/Q)/(\Delta P/P)$$

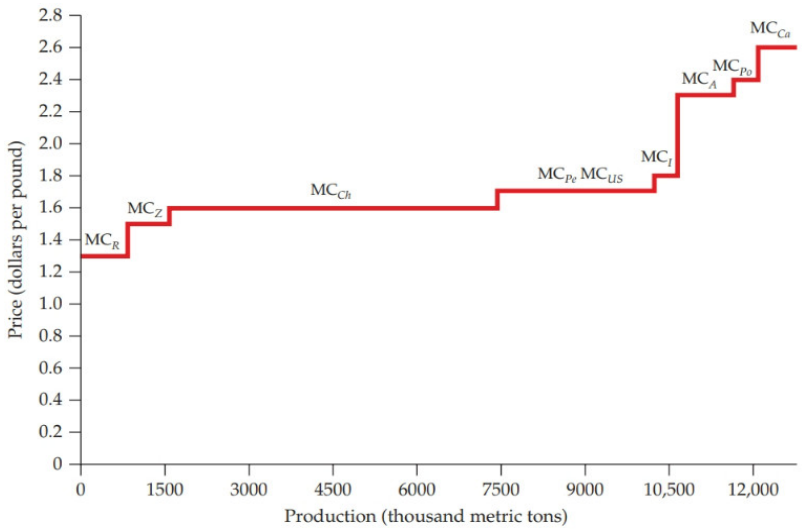
Example: Short-Run World Supply of Copper

Costs of mining, smelting, and refining copper differ because of differences in labor and transportation costs and because of differences in the copper content of the ore.

TABLE 8.1	THE WORLD COPPER INDUSTRY (2014)	
COUNTRY	ANNUAL PRODUCTION (THOUSAND METRIC TONS)	MARGINAL COST (DOLLARS PER POUND)
Australia	1000	2.30
Canada	680	2.60
Chile	5800	1.60
Indonesia	400	1.80
Peru	1400	1.70
Poland	425	2.40
Russia	850	1.30
U.S.	1370	1.70
Zambia	730	1.50

Source: U.S. Geological Survey, Mineral Commodity Summaries, January 2015 (**http://minerals.usgs.gov/minerals/pubs/commodity/copper/mcs-2015-coppe.pdf**)

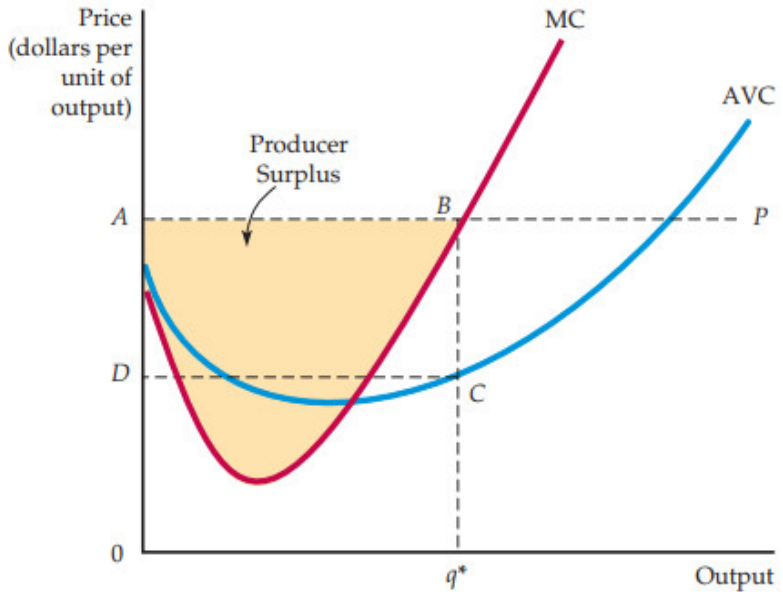
Example: Short-Run World Supply of Copper (Con't)



- The world supply curve is obtained by summing each nation's supply curve horizontally.
- At relatively low prices, the curve is quite elastic. At higher prices - say, above \$2.40 per pound - the curve becomes more inelastic.

Producer Surplus in the Short Run

producer surplus - Sum over all units produced by a firm of differences between the market price of a good and the marginal cost of production.



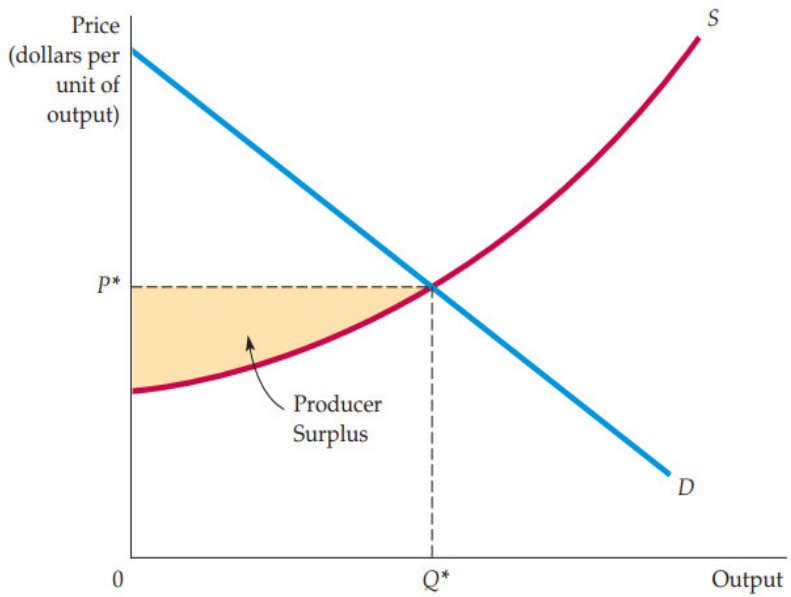
The producer surplus for a firm is measured by the yellow area, which is also equal to rectangle *ABCD*.

Producer Surplus versus Profit

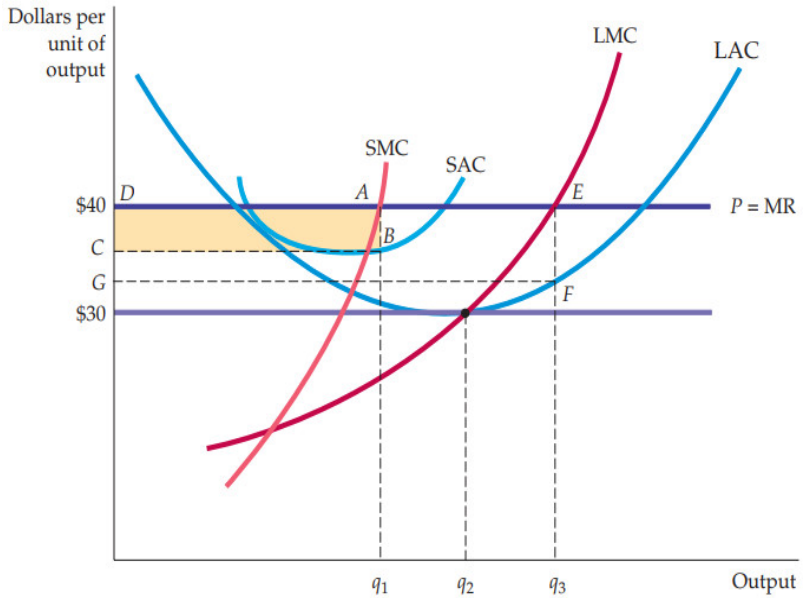
$$\text{Producer surplus} = \text{PS} = R - \text{VC}$$

$$\text{Profit} = \pi = R - \text{VC} - \text{FC}$$

Producer Surplus for a Market



Long-Run Profit Maximization



- The firm maximizes its profit by choosing the output at which price equals long-run marginal cost LMC.
- In the diagram, the firm increases its profit from *ABCD* to *EFGD* by increasing its output in the long run.

Long-Run Competitive Equilibrium

- **Accounting Profit and Economic Profit**
  - Economic profit takes into account opportunity costs. One such opportunity cost is the return to the firm's owners if their capital were used elsewhere.
  - Accounting profit equals revenues R minus labor cost wL, which is positive.
  - Economic profit π equals revenues R minus labor cost wL minus the capital cost rK:

$$\pi = R - wL - rK$$

Long-Run Competitive Equilibrium (Con't)

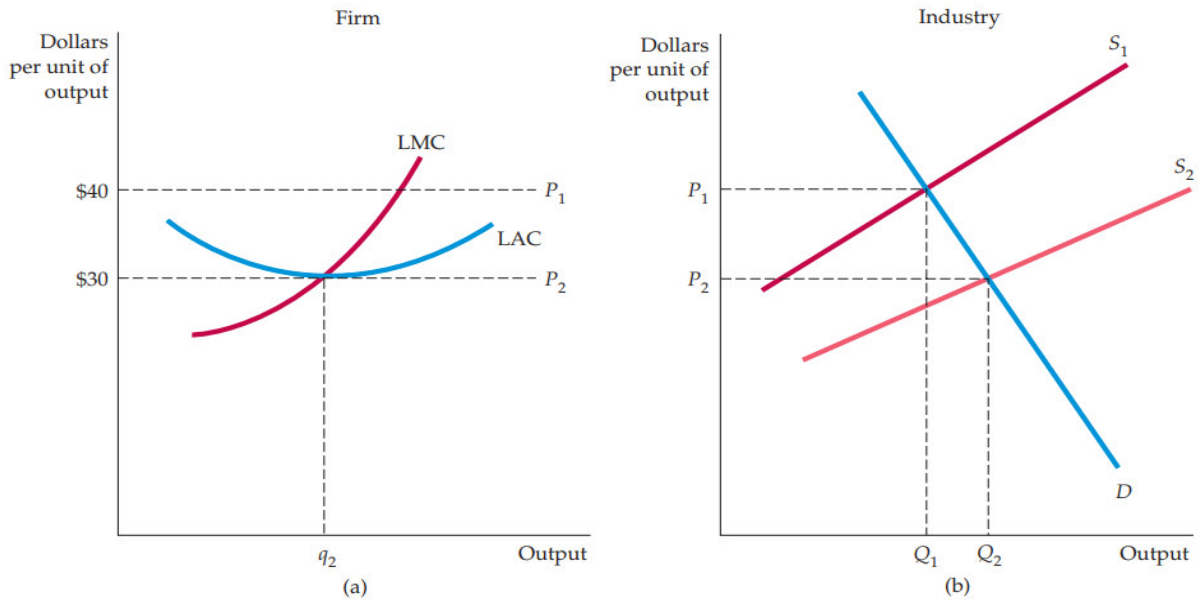
- **Zero Economic Profit** - A firm is earning a normal return on its investment—i.e., it is doing as well as it could by investing its money elsewhere.
- **Entry and Exit** - In a market with entry and exit, a firm enters when it can earn a positive long-run profit and exits when it faces the prospect of a long-run loss.
- When a firm earns zero economic profit, it has no incentive to exit the industry. Likewise, other firms have no special incentive to enter.

Long-Run Competitive Equilibrium (Con't)

A **long-run competitive equilibrium** occurs when three conditions hold:

1. All firms in the industry are maximizing profit.
2. No firm has an incentive either to enter or exit the industry because all firms are earning zero economic profit.
3. The price of the product is such that the quantity supplied by the industry is equal to the quantity demanded by consumers.

Long-Run Competitive Equilibrium

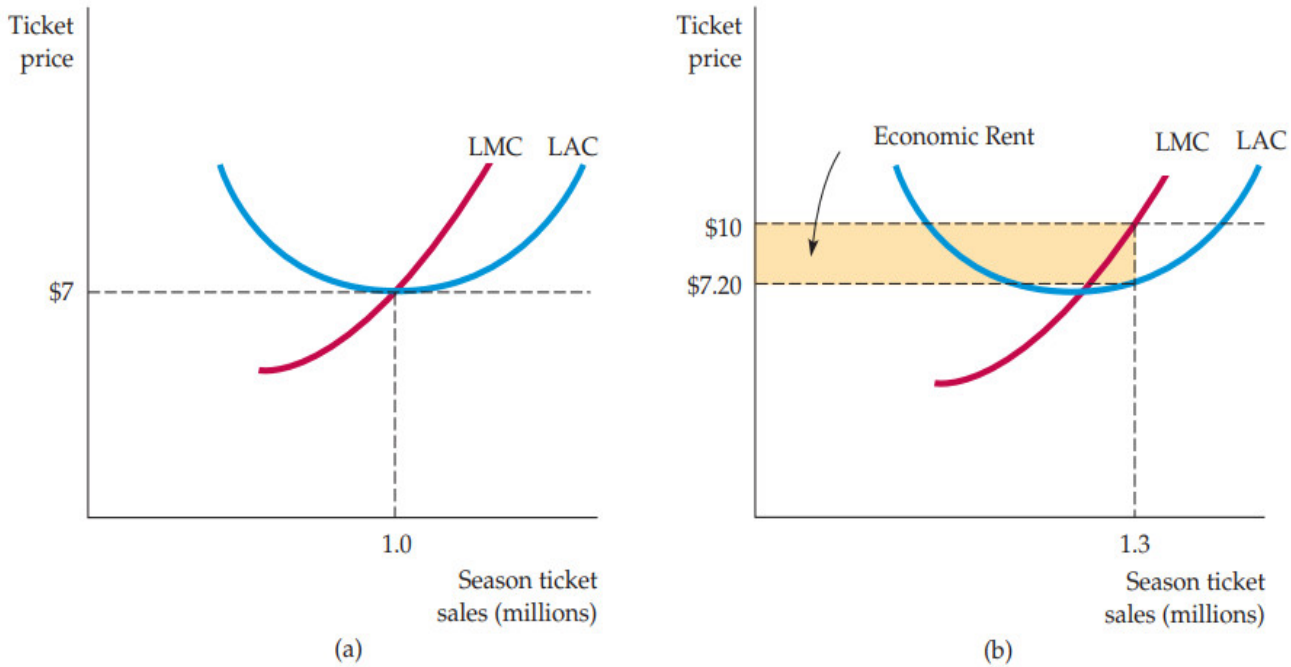


- Initially the long-run equilibrium price of a product is \$40 per unit.
- Positive profit encourages entry of new firms and causes a shift to the right of the supply curve.
- The long-run equilibrium occurs at a price of \$30, where each firm earns zero profit and there is no incentive to enter or exit the industry.

Economic Rent and Producer Surplus

- **economic rent** - Amount that firms are willing to pay for an input less the minimum amount necessary to obtain it.
  - In competitive markets, in both the short and the long run, economic rent is often positive even though profit is zero.
- **Producer Surplus in the Long Run**
  - In the long run, in a competitive market, the producer surplus that a firm earns on the output that it sells consists of the economic rent that it enjoys from all its scarce inputs.

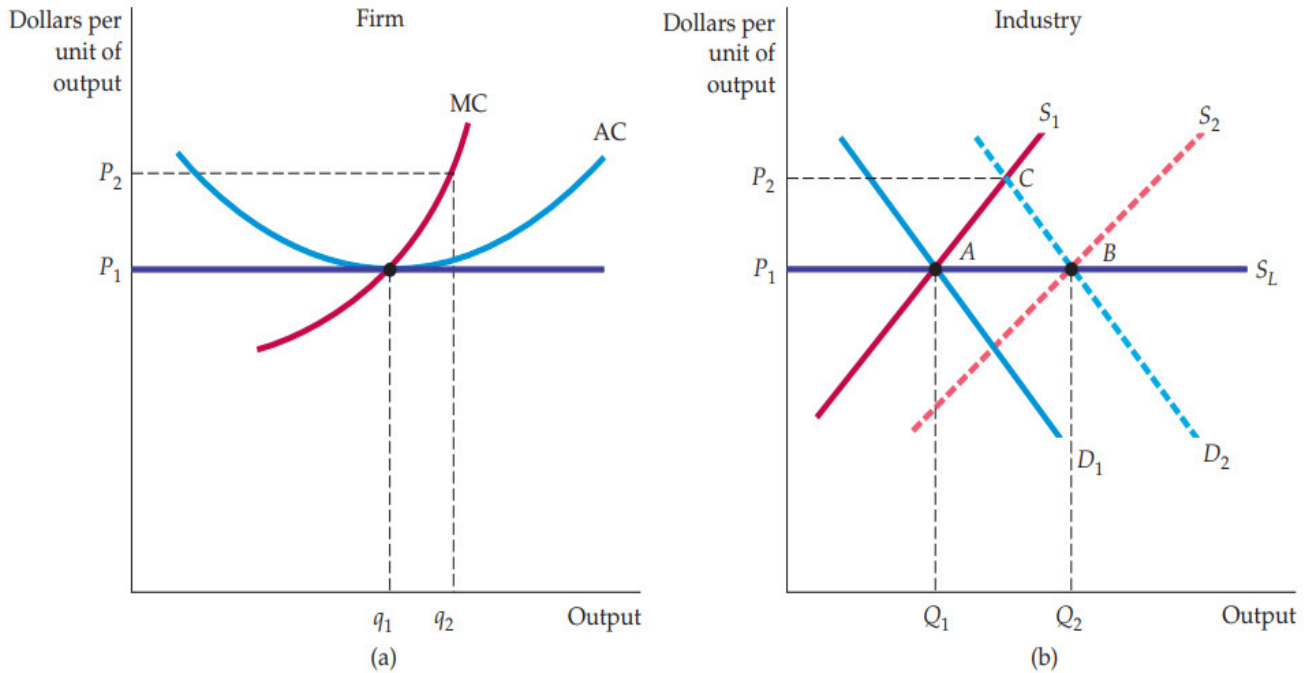
Firms Earn Zero Profit in Long-Run Equilibrium



- In long-run equilibrium, all firms earn zero economic profit.
- In (a), a baseball team in a moderate-sized city sells enough tickets so that price ($7) is equal to marginal and average cost.
- In (b), the demand is greater, so a $10 price can be charged. The team increases sales to the point at which the average cost of production plus the average economic rent is equal to the ticket price.

Constant-Cost Industry

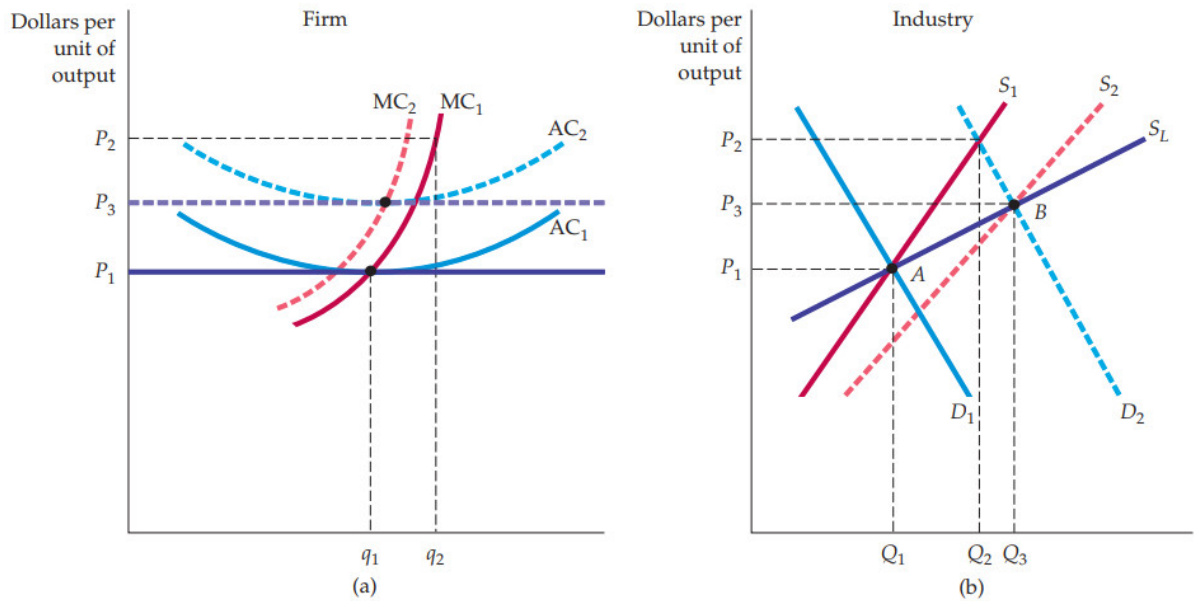
constant-cost industry - Industry whose long-run supply curve is horizontal.



The long-run supply curve for a constant-cost industry is a horizontal line at a price that is equal to the long-run minimum average cost of production.

Increasing- and Decreasing-Cost Industry

increasing-cost industry - Industry whose long-run supply curve is upward sloping.



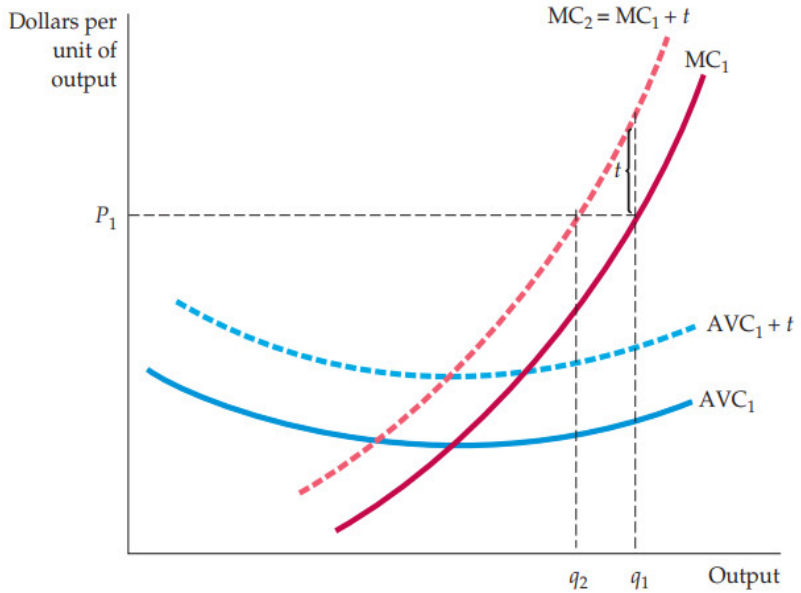
In an increasing-cost industry, the long-run industry supply curve is upward sloping.

decreasing-cost industry - Industry whose long-run supply curve is downward sloping.

Example: Constant- Increasing-, and Decreasing-Cost Industries

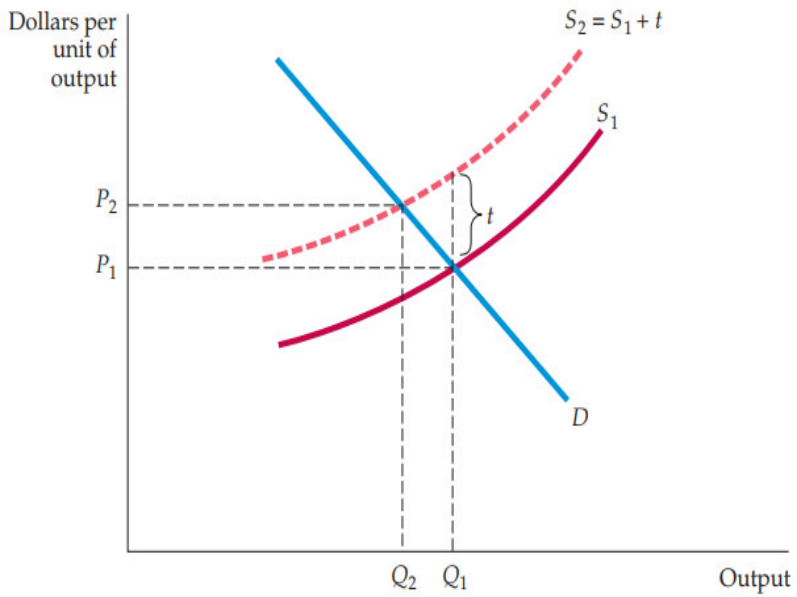
- **Coffee:** Constant-cost industry.
  - The supply of coffee is extremely elastic in the long run.
- **Oil:** Increasing-cost industry.
  - There is a limited availability of easily accessible, large volume oil fields.
- **Automobiles:** Decreasing-cost industry.
  - In the automobile industry, certain cost advantages arise because inputs can be acquired more cheaply as the volume of production increases.

The Effects of a Tax



- An output tax raises the firm's marginal cost curve by the amount of the tax.
- The firm will reduce its output to the point at which the marginal cost plus the tax is equal to the price of the product.

The Effects of a Tax (Con't)



- An output tax placed on all firms in a competitive market shifts the supply curve for the industry upward by the amount of the tax.
- This shift raises the market price of the product and lowers the total output of the industry.

Long-Run Elasticity of Supply

- The long-run elasticity of industry supply is defined in the same way as the short-run elasticity: $(\Delta Q/Q)/(\Delta P/P)$.
- In a constant-cost industry, the long-run supply curve is horizontal, and the long-run supply elasticity is infinitely large.
- In an increasing-cost industry, however, the long-run supply elasticity will be positive but finite.
- Because industries can adjust and expand in the long run, we would generally expect long-run elasticities of supply to be larger than short-run elasticities.

Summary

- Perfectly competitive markets
  - three basic assumptions
- Profit maximization
  - $MR(q) = MC(q)$
- Short-run profit maximization
  - $MC(q) = MR = P$
- Competitive firm's short-run supply curve
  - The firm's supply curve is the portion of the MC curve for which $MC > AVC$.
- Short-run market supply curve
  - The short-run industry supply curve is the summation of the supply curves of the individual firms.
  - producer surplus

Summary (Con't)

- Long-run profit maximization
  - The firm maximizes its profit by choosing the output at which price equals long-run marginal cost LMC.
  - zero economic profit
  - long-run competitive equilibrium
  - economic rent and producer surplus
- Industry's long-run supply curve
  - constant-, increase-, and decreasing-cost industries

Exercises

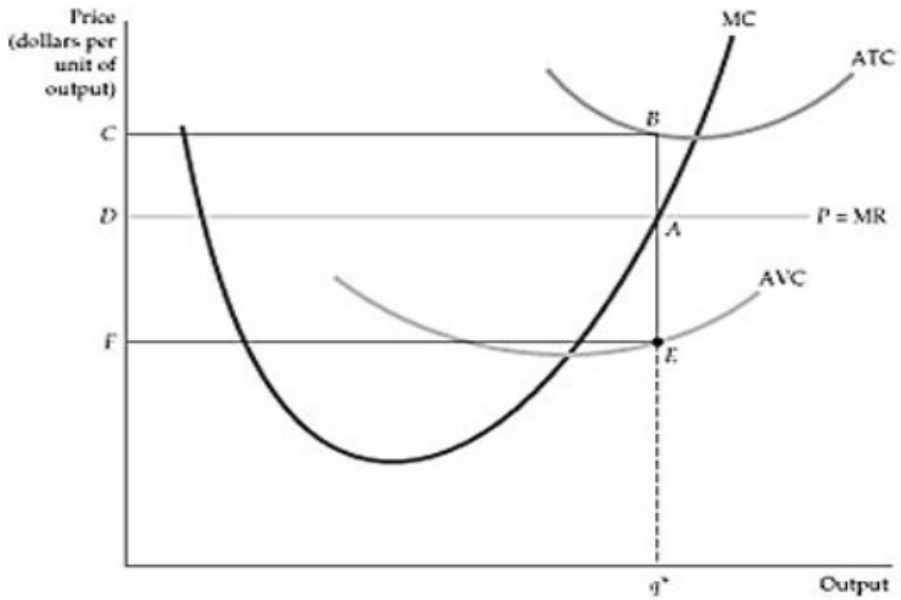
1. A few sellers may behave as if they operate in a perfectly competitive market if the market demand is:
   a. highly inelastic.
   b. very elastic.
   c. unitary elastic.
   d. composed of many small buyers.

2. At the profit-maximizing level of output, what is relationship between the total revenue (TR) and total cost (TC) curves?
   a. They must intersect, with TC cutting TR from below.
   b. They must intersect, with TC cutting TR from above.
   c. They must be tangent to teach other.
   d. They cannot be tangent to each other.
   e. They must have the same slope.

Exercises (Con't)

3. Marginal profit is negative when:
   a. marginal revenue is negative.
   b. total cost exceeds total revenue.
   c. output exceeds the profit-maximizing level.
   d. profit is negative.

Exercises (Con't)



4. Refer to the above figure. The firm in this situation should decide to
   a. shut down.
   b. produce at a loss.
   c. produce and earn the resulting profit.
   d. produce or shut down, with the same outcome.

Exercises (Con't)

5. In the short run, a perfectly competitive firm earning positive economic profit is:
   a. on the downward-sloping portion of its ATC.
   b. at the minimum of its ATC.
   c. on the upward-sloping portion of its ATC.
   d. above its ATC.
   e. below its ATC.
6. Higher input prices result in:
   a. upward shifts of MC and reductions in output.
   b. upward shifts of MC and increases in output.
   c. downward shifts of MC and reductions in output.
   d. downward shifts of MC and increases in output.
   e. increased demand for the good the input is used for.

Exercises (Con't)

7. A firm's producer surplus equals its economic profit when:
   a. average variable costs are minimized.
   b. average fixed costs are minimized.
   c. marginal costs equal marginal revenue.
   d. fixed costs are zero.
   e. total revenue equal total variable costs.

Exercises (Con't)

8. Three hundred firms supply the market for paint. For fifty of the firms, their short-run average variable costs are minimized at \$10 and short-run total costs are minimized at \$15. For the remaining firms, the short-run average variable costs and short-run average total costs are minimized at \$20 and \$25, respectively. If each firm has a U-shaped marginal cost curve then the short-run market supply curve is:
   a. U-shaped too.
   b. kinked at \$10.
   c. kinked at \$15.
   d. kinked at \$20.
   e. kinked at \$25.

Exercises (Con't)

9. Which of the following is NOT a necessary condition for long-run equilibrium under perfect competition?
   a. No firm has an incentive to enter the market.
   b. No firm has an incentive to exit the market.
   c. Prices are relatively low.
   d. Each firm earns zero economic profit.
   e. Each firm is maximizing profit.

Exercises (Con't)

10. In a constant-cost industry, an increase in demand will be followed by:
    a. no increase in supply.
    b. an increase in supply that will not change price from the higher level that occurs after the demand shift.
    c. an increase in supply that will bring price down to the level it was before the demand shift.
    d. an increase in supply that will bring price down below the level it was before the demand shift.
    e. a decrease in demand to keep price constant.